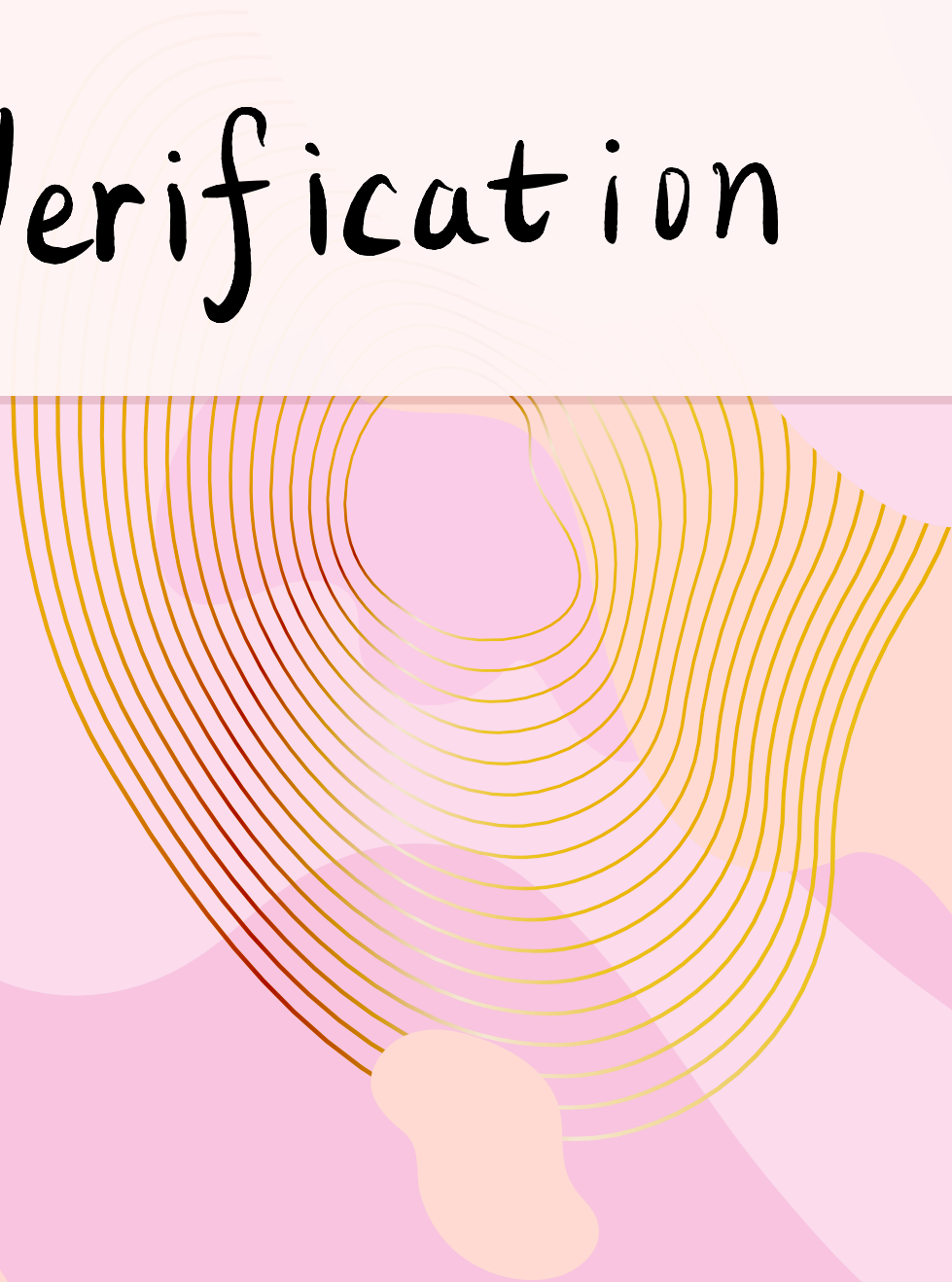


Conventions for Verification



$$S^a = \begin{matrix} x & z \\ \text{blue } \begin{bmatrix} 1 & 0 \\ a & 1 \end{bmatrix} \end{matrix}$$

$$x \rightarrow S^a \rightarrow xz^a$$

$$z \rightarrow S^a \rightarrow z$$

$$H = \begin{matrix} x & z \\ \text{blue } \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \end{matrix}$$

$$x \rightarrow H \rightarrow z$$

$$z \rightarrow H \rightarrow x^{-1}$$

$$\boxed{A_{ab}} \cong \boxed{H^{-1}} \rightarrow \boxed{S^{-a}} \rightarrow \boxed{H^{-1}} \rightarrow \boxed{S^{-1/a}} \rightarrow \boxed{H^{-1}} \rightarrow \boxed{S^{-a(b+1)}} \rightarrow \boxed{H} \rightarrow$$

In code: $A_{ab} = H * S^{-a(b+1)} * H^{-1} * S^{-1/a} * H^{-1} * S^{-a} * H^{-1}$

TEST 1:

$$H^2 = \begin{matrix} x & z \\ \text{blue } \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \end{matrix}$$

$$x \rightarrow H^2 \rightarrow x^{-1}$$

$$z \rightarrow H^2 \rightarrow z^{-1}$$

$$H^3 = \begin{matrix} x & z \\ \text{blue } \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \end{matrix}$$

$$x \rightarrow H^3 \rightarrow z^{-1}$$

$$z \rightarrow H^3 \rightarrow x$$

TEST 2: When $p=7$ $S^7 = I$

TEST 3: $HS H S H S \cong I$ $SH^2 \cong H^2 S$

TEST 4: $x^a \rightarrow H \rightarrow z^a$

$$z^a \rightarrow H \rightarrow x^{-a}$$

$$z^{-a} \rightarrow H \rightarrow x^a$$

$$\begin{matrix} x & z \\ \text{blue } \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \end{matrix} \begin{bmatrix} a \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ a \end{bmatrix}$$

$$\begin{matrix} x & z \\ \text{blue } \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \end{matrix} \begin{bmatrix} 0 \\ a \end{bmatrix} = \begin{bmatrix} -a \\ 0 \end{bmatrix}$$

$$\begin{matrix} x & z \\ \text{blue } \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \end{matrix} \begin{bmatrix} 0 \\ -a \end{bmatrix} = \begin{bmatrix} a \\ 0 \end{bmatrix}$$

TEST 5: $x^a \rightarrow S \rightarrow x^a z^a$

$$z^a \rightarrow S \rightarrow z^a$$

$$x^a z^{-a} \rightarrow S \rightarrow x^a$$

$$\begin{bmatrix} a \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} a \\ a \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ a \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ a \end{bmatrix}$$

$$\begin{bmatrix} a \\ -a \end{bmatrix}$$

$$\begin{bmatrix} a \\ 0 \end{bmatrix}$$

TEST 6: $X \xrightarrow{S^a} XZ^a$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \begin{bmatrix} 1 \\ a \end{bmatrix}$$

$Z \xrightarrow{S^a} Z$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$XZ^{-a} \xrightarrow{S^a} X$

$$\begin{bmatrix} 1 \\ -a \end{bmatrix} \quad \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

TEST 7: $X^a \xrightarrow{S^b} X^a Z^{ab}$

$$\begin{bmatrix} a \\ 0 \end{bmatrix} \quad \begin{bmatrix} a \\ ab \end{bmatrix}$$

$Z^a \xrightarrow{S^b} Z^a$

$$\begin{bmatrix} 0 \\ a \end{bmatrix} \quad \begin{bmatrix} 0 \\ a \end{bmatrix}$$

$X^a Z^{-ab} \xrightarrow{S^b} X^a$

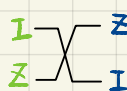
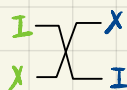
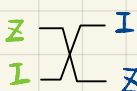
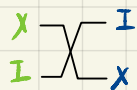
$$\begin{bmatrix} a \\ -ab \end{bmatrix} \quad \begin{bmatrix} a \\ 0 \end{bmatrix}$$

$H = \begin{matrix} X & Z \\ Z & \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \end{matrix}$

$X \xrightarrow{H} Z$

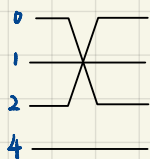
$Z \xrightarrow{H} X^{-1}$

SWAP



$$\begin{matrix} X_0 & Z_0 & X_1 & Z_1 \\ \begin{matrix} X_0 \\ Z_0 \\ X_1 \\ Z_1 \end{matrix} & \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \end{matrix}$$

SWAP(0,2,4)



$$\begin{matrix} X_0 & Z_0 & X_1 & Z_1 & X_2 & Z_2 & X_3 & Z_3 \\ \begin{matrix} X_0 \\ Z_0 \\ X_1 \\ Z_1 \\ X_2 \\ Z_2 \\ X_3 \\ Z_3 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

$$H = \begin{matrix} x & z \\ \begin{matrix} x \\ z \end{matrix} \end{matrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$x \rightarrow H \rightarrow z$$

$$z \rightarrow H \rightarrow x^{-1}$$

$$H^2 = \begin{matrix} x & z \\ \begin{matrix} x \\ z \end{matrix} \end{matrix} \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$H^3 = \begin{matrix} x & z \\ \begin{matrix} x \\ z \end{matrix} \end{matrix} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$H_1 = \begin{matrix} & x_0 & z_0 & x_1 & z_1 \\ \begin{matrix} x_0 \\ z_0 \\ x_1 \\ z_1 \end{matrix} \end{matrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$H_1^2 = \begin{matrix} & x_0 & z_0 & x_1 & z_1 \\ \begin{matrix} x_0 \\ z_0 \\ x_1 \\ z_1 \end{matrix} \end{matrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

$$H_1^3 = \begin{matrix} & x_0 & z_0 & x_1 & z_1 \\ \begin{matrix} x_0 \\ z_0 \\ x_1 \\ z_1 \end{matrix} \end{matrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

$$S^a = \begin{matrix} x & z \\ \begin{matrix} x \\ z \end{matrix} \end{matrix} \begin{bmatrix} 1 & 0 \\ a & 1 \end{bmatrix}$$

$$x \rightarrow S^a \rightarrow xz^a$$

$$z \rightarrow S^a \rightarrow z$$

$$S_1^a = \begin{matrix} & x_0 & z_0 & x_1 & z_1 \\ \begin{matrix} x_0 \\ z_0 \\ x_1 \\ z_1 \end{matrix} \end{matrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & a & 1 \end{bmatrix}$$

$$\boxed{A_{ob}} \triangleq \boxed{H^{-1}} \rightarrow \boxed{S^{1/b}} \rightarrow \boxed{H^{-1}} \rightarrow \boxed{S^b} \rightarrow \boxed{H} \quad b \in \mathbb{Z}_p^*$$

$$\frac{z^b}{x^{1/b}z} \boxed{A_{ob}} \frac{z}{x} \quad \frac{z}{x} \boxed{A_{ob}} \frac{z^{1/b}}{x^b z^{-1}} \quad \begin{matrix} x & z \\ \begin{matrix} x \\ z \end{matrix} \end{matrix} \begin{bmatrix} b & 0 \\ -1 & 1/b \end{bmatrix} \quad \checkmark$$

$$\boxed{A_{ab}} \triangleq \boxed{H^{-1}} \rightarrow \boxed{S^{-a}} \rightarrow \boxed{H^{-1}} \rightarrow \boxed{S^{-1/a}} \rightarrow \boxed{H^{-1}} \rightarrow \boxed{S^{-a(b+1)}} \rightarrow \boxed{H} \quad a \in \mathbb{Z}_p^*, b \in \mathbb{Z}_p$$

$$\frac{x^a z^b}{z^{-1/a}} \boxed{A_{ab}} \frac{z}{x} \quad \frac{z}{x} \boxed{A_{ab}} \frac{x^{-a}}{x^b z^{1/a}} \quad \begin{matrix} x & z \\ \begin{matrix} x \\ z \end{matrix} \end{matrix} \begin{bmatrix} b & -a \\ 1/a & 0 \end{bmatrix} \quad \checkmark$$